

Registration of EOSC Service Catalogues in the EOSC EU Node

Version v3, 13th of October, 2025

Abstract The EOSC EU Node (EEN) Resource Catalogue is designed to offer discovery and access to the map of resources made available in the EOSC Federation via the EOSC Nodes, including the EEN itself. In particular, it aggregates service profiles (metadata descriptions of services) collected from the service catalogues of EOSC Nodes. The resulting Catalogue is accessible via the EOSC Resource Hub discovery portal (and open APIs), and can be used to discover, access, and monitor EOSC services across the EOSC Nodes in the Federation.

This document outlines the EOSC Resource Catalogue interoperability guidelines, which EOSC Node Service Catalogue providers should follow to register their catalogues with the EEN Resource Catalogue, to make their services discoverable, visible, and accessible beyond their Node's boundaries.

For any further enquiries, refer to the online [Q&A document](#).

Editor(s):

Manghi, P. ¹

Author(s):

Manghi, P. ¹

*OpenAIRE*¹

Contributors:

Leonardo Candela (ISTI-CNR, Italy), **BlueCloud Node/IT Node**

Jessica Parland-von Essen (CSC, Finland), **FI Node**

Bob Jones (CERN, Switzerland), **EOSC-A**

Maurice Vanderfeesten (SURF, The Netherlands), **SURF Node**

EOSC-A reviewers:

TBC

Table of Contents

The EOSC EU Node Resource Catalogue	3
Integrating Service Catalogues in the EEN Resource Catalogue	5
Compliance of EOSC Service Catalogues with EEN Interoperability Guidelines	6
Registration of EOSC Service Catalogues in the EEN Resource Catalogue	6
The Contributor initiates the EEN Contributor's Dashboard self-service for enrolling the EOSC Node and the onboarding of research product catalogue(s) as a service catalogue. Detailed instructions are provided in Appendix C of this document.	6
Appendix A: APIs	7
Appendix B Schema	8
Service Bundle	8
Service	8
Controlled vocabularies	12
Example	12
Appendix C Registering your service catalogue with the EEN	15
1. Enrolment of a Node in the Federation	15
1.1 Manage Contributors	15
1.2 Basic Information	16
1.3 Profile	17
1.4 Location	18
1.5 Contact	19
1.6 Users	20
1.7 Overview	21
2. Onboard a Service Catalogue from a Node	22
2.1 Node Dashboard	22
2.2 Basic Metadata	23
2.3 Marketing	24
2.4 Classification	25
2.5 Location	26
2.6 Contact	27
2.7 Maturity	28
2.8 Management (I/II)	29
2.9 Management (II/II)	30
2.10 Data Source Information	31
2.11 Data Source Metadata (I/IV)	32
2.12 Data Source Metadata (II/IV)	33
2.13 Data Source Metadata (III/IV)	34
2.14 Data Source Metadata (IV/IV)	35
2.15 Acknowledgment and Acceptance	36
2.16 Overview	37

The EOSC EU Node Resource Catalogue

The *EOSC EU Node Resource Catalogue* (*EEN Resource Catalogue*, also known as the *EOSC Knowledge Graph*) is an indexed collection of metadata profiles that describe *EOSC Resources*. Such profiles are collected from a list of trusted sources and are today searchable and accessible via the [EOSC Resource Hub discovery portal](https://open-science-cloud.ec.europa.eu/resources) shown in Figure 1.

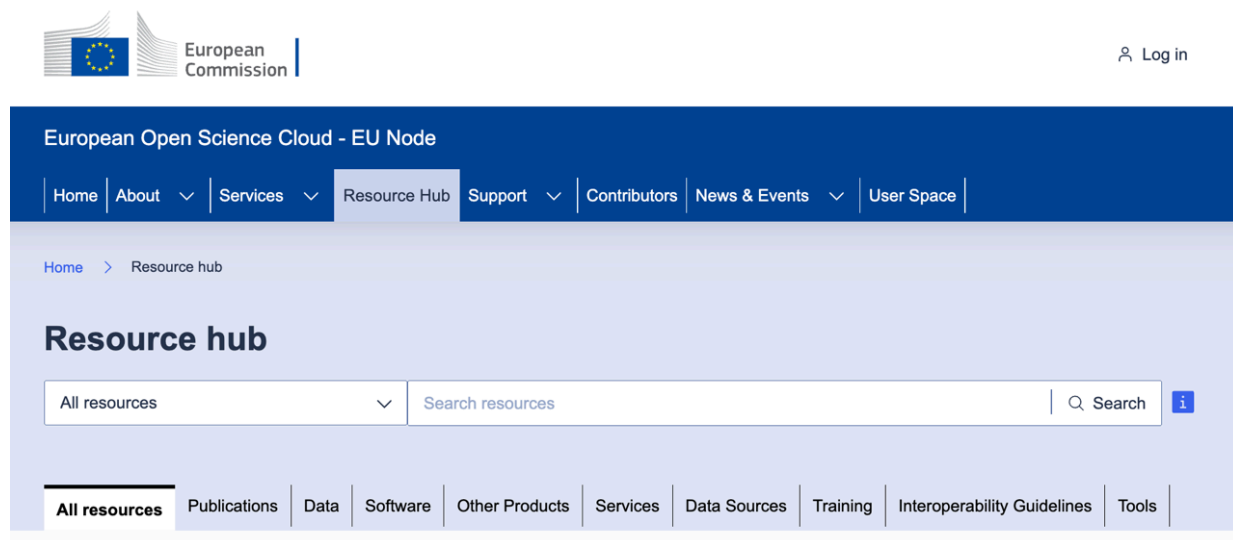


Figure 1 - EOSC Resource Hub discovery portal (<https://open-science-cloud.ec.europa.eu/resources>).

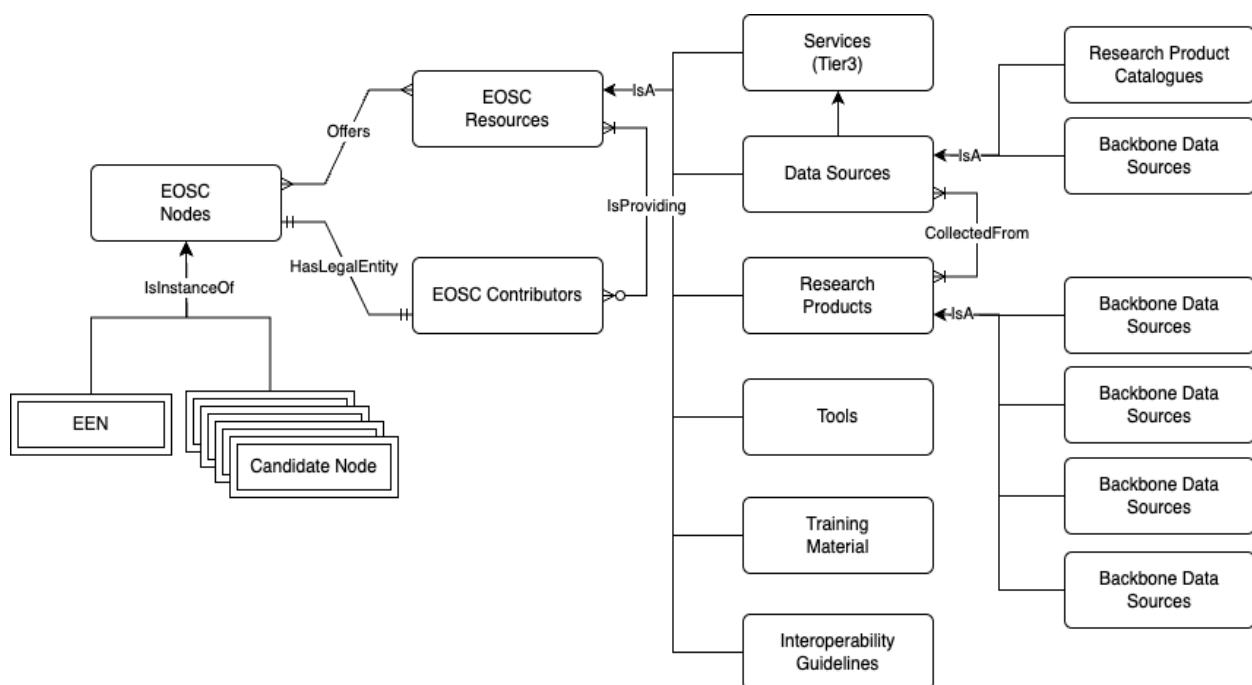


Figure 2 - EEN Resource Catalogue data model.

The purpose of the EEN Catalogue is to offer to users and machines (via APIs) a single discovery entry-point to all resources made available by EOSC Nodes (including the EEN itself) through the EOSC Federation. Its resource data model is shown in Figure 2. All resources in the catalogue have at least one Contributor (an organization that provides or operates the resource) and are offered via at least one EOSC Node, which could be the EEN or one of the candidate Nodes in the build-up phase. An EOSC Node has one responsible Contributor, namely the legal entity organization entitled to its governance.

This document focuses on the processes and interoperability guidelines that developers of EOSC Node service catalogues must follow and implement to register and federate their catalogues and register to the EEN Resource Catalogue the service metadata profiles they include.

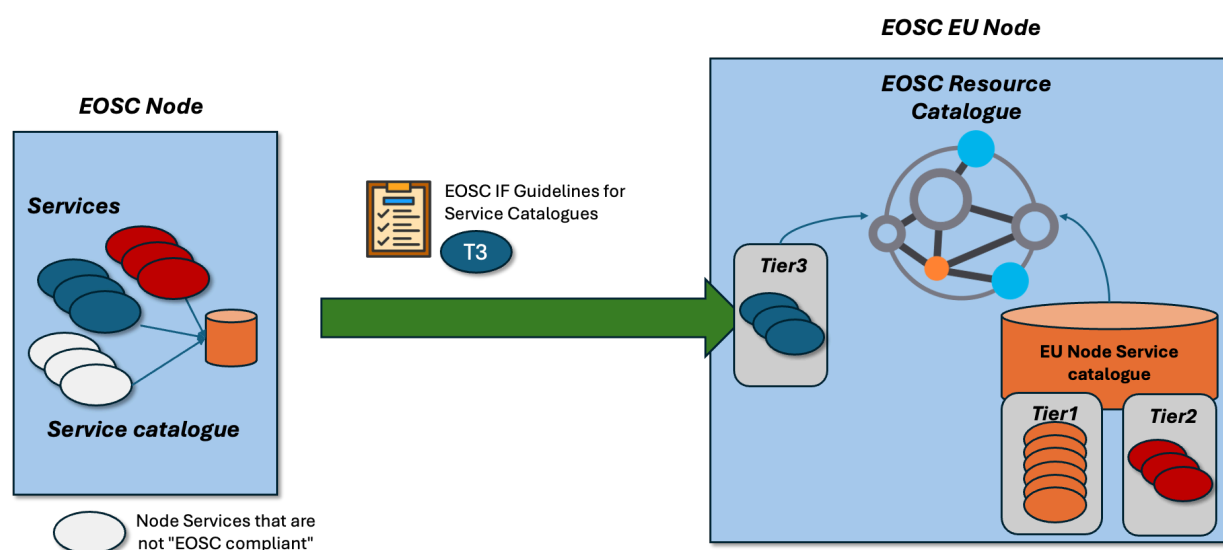


Figure 3 - EOSC Node Service Catalogues: onboarding Tier 3 services

EOSC Node Service catalogues

According to the EOSC-A, an EOSC service is a **digital service** delivering capabilities via web user interfaces and/or APIs: a service is not software, but rather software in execution; a service is not a website providing information about an office, organization, or similar. The EOSC-A has not yet defined a service classification vocabulary. Such a classification is deemed necessary and will be collaboratively defined in the next stage of the build-up phase by the EOSC-A subgroup of resource catalogues. Examples of services are:

- **Research Product catalogues/data sources** (generalist and thematic repositories, data and software repositories, aggregators, CRIS systems, etc.);
- **Service catalogues** (catalogues of service descriptors);
- **PID registries** (e.g., organization and author national PID systems);

- **Scientific data collections** (timeseries database, observational/sensor data, curated data collections, e.g., Copernicus, OpenCitations, OpenAIRE Graph, etc.);
- **Data management** (DMPs, FAIR validation, validators, policies, rules, IPR management)
- **Research monitoring** (research assessment, impact, trends, dashboards);
- **Computing resource provision** (access to computing resources, cloud, HPC, etc.);
- **Storage resource provision** (access to storage resources, distributed file systems, S3, workspace, DropBox-like, B2DROP, etc.);
- **Training services** (online courses, certifications, badges);
- **Data analytics** (data processing, data conversion, RStudio, KNIME, workflow engines...);
- **Research support** (collaborative editing of papers, social networking, conference support, e.g., Overleaf, Indico);
- **Science Gateways, i.e., entry points to a package of services** (e.g., Virtual Research Environments, collaboratories, user spaces).

A Service Catalogue enables EOSC Node service providers to describe their service offerings using a **service metadata profile**. Service catalogues are free to implement service profile information models tailored to their specific application scenarios, providing a single entry point for the Node's portfolio of services.

An EOSC Node can maintain only one service catalogue. Once a catalogue is registered and validated, its service metadata profiles are aggregated into the EOSC Resource Catalogue, including provenance information about the EOSC Node through which the services are offered. The EOSC EU Node defines three tiers of services:

- **Tier 1:** services owned by the EC (procured/provisioned), fully integrated in the EEN service offer, orderable via the User Space with credits;
- **Tier 2:** services owned by third-party providers via EOSC Nodes, integrated in the EEN service portfolio, and orderable via the EEN User Space with credits;
- **Tier 3:** services in the service catalogue of recognized EOSC Nodes (candidates), whose profiles are collected by the EEN Service Catalogue to offer cross-Node discoverability and access.

EOSC Node Service catalogues must onboard to the EEN Resource Catalogue only Tier 3 service profiles, enabling their discovery via catalogue entries that offer an external link to the service. Due to their “ENN-embedded” nature, Tier 2 service profiles are directly registered to the EOSC EU Node service catalogue and do not need to be exposed by the related Service Catalogue. Figure 3 shows a scenario where an EOSC EU Node service catalogue offers a portfolio of services where some are not EOSC-compliant (white circles), e.g., can only be accessed by local users, some are Tier 2 integrated to the EOSC EU Node (red circles), and some are Tier 3 compatible (blue circles).

Integrating Service Catalogues in the EEN Resource Catalogue

This document describes the EEN interoperability guidelines and tools that Contributors of EOSC Nodes must implement to register their service catalogue and share their service profiles via the EEN Resource Catalogue, thereby enabling discovery and monitoring of resources across Nodes in the EOSC Federation. To this aim, as illustrated in Figure 3, EOSC Contributors, responsible for the Node catalogues, will:

- **Make their catalogue compliant** with the EEN interoperability guidelines for service

catalogues defined in Appendix A (API protocol) and Appendix B (service profile metadata format);

- **Register the catalogues** in the EEN Resource Catalogue;
- **Validate the catalogues** to verify technical compliance.

The EEN Resource Catalogue will **aggregate** (“pull”) service metadata records from the catalogues at regular intervals.

In the following sections, such steps are described in detail.

Compliance of EOSC Service Catalogues with EEN Interoperability Guidelines

To comply with the EEN Resource Catalogue interoperability guidelines, Service catalogues must implement the following interoperability guidelines:

- **APIs:** GET operation of the Service Controller (**Appendix A**);
- **Exchange format:** Service Bundle schema (**Appendix B**), by appropriately crosswalking from their local service metadata schema to the “service bundle” schema.

If the EOSC Node has no service catalogue, solutions are recommended in **Appendix D**.

Registration of EOSC Service Catalogues in the EEN Resource Catalogue

The Contributor initiates the EEN Contributor’s Dashboard self-service for enrolling the **EOSC Node** and the onboarding of research product catalogue(s) as a **service catalogue**. Detailed instructions are provided in **Appendix C** of this document.

Appendix A: APIs

The API of the service catalogue must implement one method

```
/services    //Get a list of Service profiles based on a set of filters.
```

Params:

- keyword: String (Keyword to refine the search) [optional]
- from: String (Starting index in the result set, default 0) [optional]
- quantity: String (Quantity to be fetched, default 10) [optional]
- order: String (Order of results - asc/desc, default asc) [optional]
- sort: String (Field to use for ordering) [optional]

Example:

- resource-catalogue-url/services?quantity=50

The response format will look like:

```
{
  "total": 87,
  "from": 0,
  "to": 50,
  "results":
  [
    {
      "id": "21.11161/zNsT3H",
      "service":
      {
        "id": "21.11161/zNsT3H",
        "name": "Research Data Unipd",
        ...
      }
    },
    ...
  ]
}
```

Appendix B Schema

For a technical specification of the schema, refer to:

<https://code-repo.d4science.org/giambattista.bloisi/een-catalog-specs>

- **python response model:**
https://code-repo.d4science.org/giambattista.bloisi/een-catalog-specs/src/branch/main/catalogue_model.py
- **end-point sample implementation (mocking data from SURF):**
<https://code-repo.d4science.org/giambattista.bloisi/een-catalog-specs/src/branch/main/main.py>
- **JSON schema:**
https://code-repo.d4science.org/giambattista.bloisi/een-catalog-specs/src/branch/main/service_catalogue.schema.json

Metadata service profile:

Service Bundle

Id	Field	Type	Required	Multiplicity	Description	Example
1	id	String	Yes	1	Unique identifier for the service	surf-node:servicebundle:123456 (It can be a local unique ID, or a PID)
2	service	<Service>	Yes	1	Metadata of the actual resource.	

Service

Id	Field	Type	Required	Multiplicity	Description	Example
1	id	String	Yes	1	Unique identifier for the service (<u>use the same value as the Service Bundle ID</u>)	surf-node:servicebundle:123456

Id	Field	Type	Required	Multiplicity	Description	Example
2	alternativeIdentifiers	List	No	0..1	List of alternative identifiers for the service.	
2.1	type	String	No	0..1	Type of the alternative identifier.	OpenDOAR, re3data, hd1, doi, FAIRSharing
2.2	value	String	No	0..1	Value of the alternative identifier.	10.1234/example.svc.123456, 21.1234/EX-SVC-123456 When using PIDs for instruments https://docs.pidinstitute.org/en/latest/epic-cookbook/handles.html
3	abbreviation	String	No	0..1	Abbreviation of the service's name.	EXSVC
4	name	String	Yes	1	Full name of the service.	Example Service for Research Data Processing
5	webpage	URL	Yes	1	URL of the service's webpage.	https://service.example.org
6	description	String	Yes	1	An abstract with a detailed description of the service.	Example Service for Research Data Processing provides scalable compute, curated datasets, and workflows to support reproducible research across multiple scientific domains.

Id	Field	Type	Required	Multiplicity	Description	Example
7	tagline	String [max 80 char]	No	0..1	Short tagline summarizing the service.	Scalable, reproducible research processing
8	logo	URL	No	0..1	URL of the service's logo.	https://service.example.org/assets/logo.png
9	scientificDomains	List	Yes	1	List of scientific domains related to the service.	
9.1	scientificDomain	Controlled Vocabulary: SCIENTIFIC_DOMAIN	Yes	1	Scientific domain related to the catalogue.	scientific_domain-social_sciences, scientific_domain-engineering_and_technology
10	categories	List	Yes	1	Categories the service, i.e., type of service	
10.1	category	Controlled Vocabulary: CATEGORY	Yes	1	Category of the service	category-processing_and_analysis-data_analysis, category-access_physical_and_infrastructures-compute
11	targetUsers	Controlled Vocabulary: TARGET_USERS	Yes	1..N	List of target users for the service	target_user-researchers, target_user-research_groups, target_user-research_infrastructure_managers, target_user-providersProviders
12	accessModes	List of Controlled Vocabulary:	No	0..N	Types of access provided by the service	[access_mode-free_conditionally]

Id	Field	Type	Required	Multiplicity	Description	Example
		ACCESS_MODE				
13	tags	List String	No	0..N	Tags associated with the service.	[Reproducibility, workflows, datasets, high-performance]
14	languageAvailabilities	List of String	Yes	1..N	List of language availabilities of the service (ISO 639-1 Code)	[en,nl]
15	helpdeskEmail	String	No	0..1	Email address for the service's helpdesk.	helpdesk@example.org
16	securityContactEmail	String	No	0..1	Email address for security contact.	security@example.org
17	trl	Controlled Vocabulary: TRL	Yes	1	Technology Readiness Level of the service.	Tr1-8
18	userManual	URL	No	0..1	URL of the user manual.	https://service.example.org/docs/user-manual
19	termsOfUse	URL	No	0..1	URL of the terms of use.	https://service.example.org/legal/terms
20	privacyPolicy	URL	No	0..1	URL of the privacy policy.	https://service.example.org/legal/privacy
21	accessPolicy	URL	No	0..1	URL of the access policy.	https://service.example.org/legal/access
22	orderType	Controlled Vocabulary: ORDER_TYPE	Yes	1	Service access modality	order_type-open_access, order_type-order_required

Controlled vocabularies

CATEGORY

As defined in:

<https://github.com/EOSC-Lot-1/een-resource-catalogue-docs/blob/eosc/vocabularies/CATEGORY.json>

SCIENTIFIC_DOMAIN

As defined in:

https://github.com/EOSC-Lot-1/een-resource-catalogue-docs/blob/eosc/vocabularies/SCIENTIFIC_DOMAIN.json

ACCESS_MODE

As defined in:

https://github.com/EOSC-Lot-1/een-resource-catalogue-docs/blob/eosc/vocabularies/ACCESS_MODE.json

TARGET_USERS

As defined in:

https://github.com/EOSC-Lot-1/een-resource-catalogue-docs/blob/eosc/vocabularies/TARGET_USERS.json

TRL (technology readiness levels)

As defined in:

<https://github.com/EOSC-Lot-1/een-resource-catalogue-docs/blob/eosc/vocabularies/TRL.json>

ORDER_TYPE

As defined in :

https://github.com/madgeek-arc/resource-catalogue-docs/blob/master/vocabularies/ORDER_TYPE.json

Example

Note: for a full API response see Appendix A.

```
{
  "id": "surf-node:servicebundle:123456",
  "service": {
    "id": "surf-node:servicebundle:123456",
    "alternativeIdentifiers": [
```

```

    {
      "type": "doi",
      "value": "10.1234/example.svc.123456"
    },
    {
      "type": "hdl",
      "value": "21.1234/EX-SVC-123456"
    }
  ],
  "abbreviation": "EXSVC",
  "name": "Example Service for Research Data Processing",
  "webpage": "https://service.example.org",
  "description": "Example Service for Research Data Processing provides
scalable compute, curated datasets, and workflows to support reproducible
research across multiple scientific domains.",
  "tagline": "Scalable, reproducible research processing",
  "logo": "https://service.example.org/assets/logo.png",
  "scientificDomains": [
    {
      "scientificDomain": "scientific_domain-social_sciences"
    },
    {
      "scientificDomain": "scientific_domain-engineering_and_technology"
    }
  ],
  "categories": [
    {
      "category": "category-processing_and_analysis-data_analysis"
    },
    {
      "category": "category-access_physical_and_eInfrastructures-compute"
    }
  ],
  "targetUsers": [
    "target_user-researchers",
    "target_user-research_groups",
    "target_user-research_infrastructure_managers",
    "target_user-providers"
  ],
  "accessModes": ["access_mode-free_conditionally"],
  "tags": [
    "Reproducibility",
    "workflows",
    "datasets",
    "high-performance"
  ],
  "languageAvailabilities": [
    "EN",
    "NL"
  ],
  "helpdeskEmail": "helpdesk@example.org",
  "securityContactEmail": "security@example.org",
  "trl": "trl-8",
  "userManual": "https://service.example.org/docs/user-manual",
  "termsOfUse": "https://service.example.org/legal/terms",
  "privacyPolicy": "https://service.example.org/legal/privacy",
  "accessPolicy": "https://service.example.org/legal/access",
  "orderType": "order_type-open_access"
}

```


Appendix C Registering your service catalogue with the EEN

To be able to federate your service catalogue(s) you need to register them as data sources under your node. This is done in the EEN contributor's dashboard. If you already have enrolled your node, move straight to section 2.

1. Enrolment of a Node in the Federation

1.1 Manage Contributors

The screenshot displays the 'Manage Contributors' interface. On the left is a sidebar with the European Commission logo, 'EOSC EU Node', 'Resource Hub', user information for 'Diana Brown', and a 'Logout' button. Below this are sections for 'SERVICES' (File Sync & Share, Interactive Notebooks, Large File Transfer, Cloud Container Platform, Virtual Machines, Bulk Data Transfer, Other Services) and 'GENERAL' (Groups, Orders, Credits, Manage Contributors, Favourites). The 'Manage Contributors' link is highlighted with a red circle labeled '1'. The main area is titled 'Manage Contributors' with the subtitle 'View and manage your contributor organisations.' It includes an 'Add Contributor +' button and a 'Show:' filter section with checkboxes for 'Onboarded', 'Enrolled', 'Pending', 'Rejected', 'Draft' (checked), 'Requires improvement', and 'Offboarded'. This filter section is highlighted with a red circle labeled '2'. Below the filters, a 'Draft (5)' section shows a list of draft contributors and nodes. The first row contains three entries: 'Test Contributor 14/02/2025 1:40', 'Test Contributor 14/02/2025 10:45 - updated', and 'Test'. The second row contains two entries: 'Test Node 23/7/2025 No 3' and 'Test Node 23/7/2025 No 2'. The 'Test Node 23/7/2025 No 3' entry is highlighted with a red circle labeled '3'. The 'Test Node 23/7/2025 No 2' entry has a red circle labeled '4' pointing to its three vertical menu lines.

- Once the EOSC EU Node Onboarding Team creates your Node draft entry, you will receive an email to finish enrolment. To do so, as a first step, you may log in to your **User Space** and navigate to the **Manage Contributors** page - (1)
- To facilitate the detection of the draft Node entry, you may use the filters on the top of the page - (2)
- To resume and complete the Node enrolment procedure, you may either click on the draft Node's name, or on the three vertical lines and select the **Resume Application** option - (3), (4)

1.2 Basic Information

Edit Node

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

1 **Basic Information *** 2* 1*

2 **Profile *** 2*

3 **Location *** 4*

4 **Contact *** 3*

5 **Users**

6 **Overview**

0/4 mandatory steps completed
23%

Cancel and Exit Update Node

Name *
Name of the Contributor organisation. In case it is an individual, their full name could be provided.
Test Node 23/7/2025 No 3
Up to 100 characters

Abbreviation *
Abbreviation or short name of the Contributor.
TN
Up to 30 characters

Website *
The website address of the Contributor.
use http:// or https://

Legal Entity *
Whether the Contributor is a legal entity or not.
☒ No ☐ Yes

Node Type
The type of the Node.

Next >

- After entering the Node enrolment wizard, you will be presented with a two-pane view. The various steps of the wizard are enumerated on the left pane. Each step has a number of mandatory fields marked with a circle next to its title. The total percentage of mandatory fields completed for the entire wizard is also indicated at the bottom of the left pane - (1), (2), (3)
- In the first step of the wizard, you will be asked to fill-in some **Basic Information** for your Node, including its **Name**, **Abbreviation** and **Website** address (name and abbreviation follow the agreed federation node naming conventions) - (4), (5), (6)
- You must also indicate whether the Node is a **Legal Entity** or not - (7)
- Finally, you may also optionally select the type of the Node (i.e., one of **EU Wide**, **National**, **Regional**, or **Thematic**) - (8)
- To proceed to the next step of the wizard, click on the **Next** button - (9)
- You may cancel the process and exit at any time without saving by clicking on the **Cancel and Exit** button - (10)
- To update the draft application for your Node enrolment and exit the wizard without submitting it, click on the **Update Node** button - (11)

1.3 Profile

The screenshot shows the 'Edit Node' interface for a user named Diana Brown. The left sidebar contains navigation links for Overview, Notifications, Tools Hub, SERVICES (File Sync & Share, Interactive Notebooks, Large File Transfer, Cloud Container Platform, Virtual Machines, Bulk Data Transfer, Other Services), GENERAL (Groups, Orders, Credits, Manage Contributors, Favourites), and Logout. The main content area is titled 'Edit Node' and includes a progress bar showing '1/4 required steps completed' at 30%. The 'Profile' step is highlighted, with a list of steps: 1. Basic information, 2. Profile, 3. Location, 4. Contact, 5. Users, and 6. Overview. The 'Description' field (1) is a large text area with a placeholder 'A high-level, non-... of the Contributor.' and a note 'Up to 1000 characters'. The 'Logo' field (2) is a text area with a placeholder 'A URL that directs to the logo of the Contributor.' and a note 'use http:// or https://'. The bottom of the page has 'Cancel and Exit' and 'Update Node' buttons, and a 'Previous' and 'Next' navigation bar.

- At the **Profile** step of the wizard, you must provide a short **Description** of your Node organization (i.e., not more than 1000 characters).
- You must also provide a URL pointing to the **Logo** of your Node organization. The logo can be of any image type (e.g., PNG, JPEG) - (1), (2)

1.4 Location

Edit Node

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

Basic information * ✓
Profile * ✓
3 Location * 4*
4 Contact * 3*
5 Users
6 Overview

2/4 required steps completed
46%

Cancel and Exit Update Node

Address *
Street name and number of the Contributor's premises.
Up to 50 characters

Postal Code *
The postal code of the Contributor's premises' address.

City *
The city where the Contributor's premises are located.

Region
The region where the Contributor's premises are located.
Up to 20 characters

Country *
The country where the Contributor's premises are located.

Previous Next >

- At the **Location** step of the wizard, you must provide the location information of your Node organization. Specifically, you must provide the **Address**, **Postal Code**, **City**, and **Country** of your organization - (1), (2), (3), (4)
- Optionally, you may provide the **Region** of your Node Organisation - (5)

1.5 Contact

Edit Node

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

- Basic information *
- Profile *
- Location *
- 4 Contact *** (3*)
- 5 Users
- 6 Overview

3/4 required steps completed
75%

Cancel and Exit Update Node

Main Contact *

1 First Name *
The first name of the Contributor's main contact person.
Up to 50 characters

2 Last Name *
The last name of the Contributor's main contact person.
Up to 50 characters

3 Email *
The email of the Contributor's main contact person.

4 Phone Country Code
The phone country code

4 Phone
The phone of the Contributor's main contact person.
Up to 10 characters

5 Position
The organisational position of the Contributor's main contact person.
Up to 50 characters

Previous Next >

- Next, at the **Contact** step of the wizard, you must provide the details of the Node's main contact person, namely the Node Coordinator. Specifically, you must provide his/her **First Name**, **Last Name** and **Email**. Please note that the main contact must be a registered EOSC EU Node user - (1), (2), (3)
- Optionally, you may provide his/her **Phone** number and organizational **Position** - (4), (5)

1.6 Users

The screenshot displays the 'Edit Node' wizard in the EOSC EU Node interface. The left sidebar contains navigation links for 'Overview', 'Notifications', 'Tools Hub', 'SERVICES' (File Sync & Share, Interactive Notebooks, Large File Transfer, Cloud Container Platform, Virtual Machines, Bulk Data Transfer, Other Services), 'GENERAL' (Groups, Orders, Credits), 'Manage Contributors', and 'Favourites'. The main content area is titled 'Edit Node' and shows a progress bar at the bottom indicating '4/4 required steps completed' and '100%'. The 'Users' step is active, showing a form with three fields: 'First name' (Diana), 'Last name' (Brown), and 'Email' (brown_03@example.domain). Each field is highlighted with a red circle and a number (1, 2, 3 respectively). The 'Users' step is the fifth step in the wizard, as indicated by the '5' in the step indicator.

- At the **Users** step of the wizard, you are presented with non-editable fields containing the details of the administrator user of the Node under enrolment, which was provided during the initial creation of the draft application by the EOSC EU Node Onboarding Team. Specifically, you can view his/her **First Name**, **Last Name** and **Email**. The administrator user must have an active account on the EOSC EU Node - (1), (2), (3)

1.7 Overview

Edit Node

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

- Basic information *
- Profile *
- Location *
- Contact *

5 Users

6 **Overview**

4/4 required steps completed 100%

Cancel and Exit Update Node

Basic Edit

Name: Test Node 23/7/2025 No 2

Abbreviation: TN

Legal Entity: No

Website: https://test.com

Node Type:

OpenAIRE Community Tag:

Profile Edit

Description: This is a Node description

Logo: https://github.com/test-images/png/blob/main/202105/cs-black-000.png

Location Edit

Address: Test Address 12

Postal Code: 12345

City: Athens

Region: Attica

Country: Greece

Contact Edit

Main Contact:

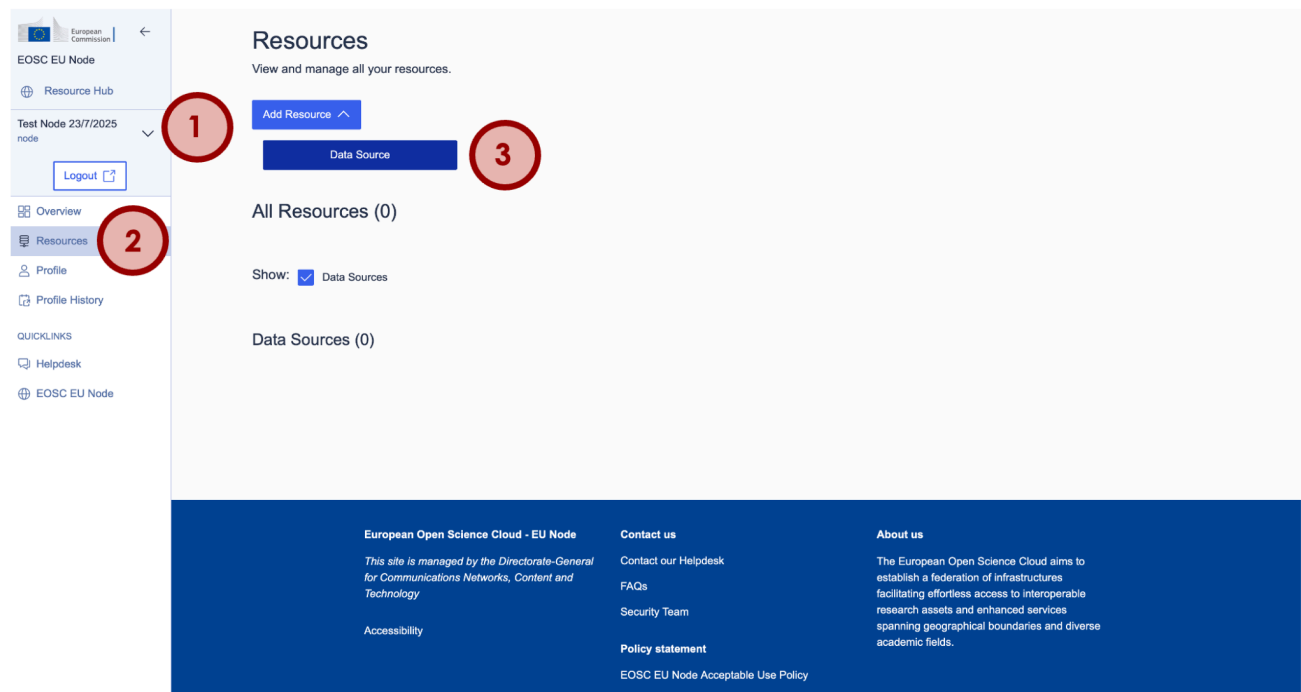
First Name: Georgios

Previous Publish Node

- Finally, at the **Overview** step of the wizard, you have the opportunity to go through all the details that you have provided for your Node organisation
- To edit a field before submitting, you may either click on the corresponding step title at the left pane of the wizard, or at the **Edit** button above each category - (1), (2)
- To submit the application for review, click on the Publish Node button. You will be presented with a confirmation modal. Upon successful submission the Node's application will appear as having the 'Pending' status in the **Manage Contributors** tab in your **User Space**. Once the application is validated and accepted by the EOSC EU Node Onboarding Team and the European Commission, your Node organisation will appear as 'Enroled' in the **Manage Contributors** tab in your **User Space** - (3)

2. Onboard a Service Catalogue from a Node

2.1 Node Dashboard



- When you submit your Node enrolment application for review, you will be able to visit your **Node Dashboard** page by clicking on the arrow next to your username while in the **User Space** and selecting your Node. Please note that if your Node enrolment application is under review by the EOSC EU Node Onboarding Team and the European Commission, the status of your node will appear as 'Pending'. From there, you may start onboarding your **Service Catalogue** even while your enrolment application is still pending - (1)
- To onboard a **Service Catalogue**, visit the **Resources** tab in the **Node Dashboard**, click on the **Add Resource** button and then on the **Data Source** button. Doing so will initiate the **Create a Data Source** wizard - (2), (3)

2.2 Basic Metadata

European Commission

EOSC EU Node

Resource Hub

Test Node 23/7/2025

node

Logout

Overview

Resources

Profile

Profile History

QUICKLINKS

Helpdesk

EOSC EU Node

Create a Data Source

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

1 Basic * 2*

2 Marketing * 2*

3 Classification * 2*

4 Location

5 Contact * 3*

6 Maturity

7 Management * 2*

8 Data Source Information * 3*

9 Data Source Metadata * 10*

10 Acknowledgement and Acceptance

11 Overview

0/7 required steps completed

0%

Cancel and Exit

Save Data Source as Draft

Next >

Name *

The name of the Data Source.

Up to 80 characters

3

Abbreviation

An abbreviation or short name of the Data Source.

Up to 20 characters

4

Webpage *

A webpage containing information about the Data Source.

use http:// or https://

5

6

7

8

- In the first page of the **Create a Data Source** wizard, you will be presented with a two-pane view. The various steps of the wizard are enumerated in the left panel. Each step has several mandatory fields marked with a circle next to its title. The total percentage of mandatory fields completed for the entire wizard is also indicated at the bottom of the left pane - (1), (2)
- During the first step of the wizard, you will be asked to fill-in some **Basic Information** for your Service Catalogue, including its **Name**, **Abbreviation**, and **Website** address - (3), (4), (5)
- You may cancel the process and exit at any time without saving by clicking on the **Cancel and Exit** button - (6)
- To update the draft application for your Service Catalogue onboarding and exit the wizard without submitting it, click on the **Save Data Source as Draft** button - (7)
- To proceed to the next step of the wizard, click on the **Next** button - (8)

2.3 Marketing

Create a Data Source

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

☒ Basic * ☒ Marketing * (1*)

3 Classification * (2*)

4 Location

5 Contact * (3*)

6 Maturity

7 Management * (2*)

8 Data Source Information * (3*)

9 Data Source Metadata * (10*)

10 Acknowledgement and Acceptance

11 Overview

1/7 required steps completed
12%

[Cancel and Exit](#) [Save Data Source as Draft](#) [Previous](#) [Next >](#)

Description *
A high-level, non-technical description of what the Data Source offers.

1

The "Description" field is required
Up to 1000 characters

Tagline
A short catchphrase for marketing and advertising purposes.

4

Logo
A link to the logo of the Data Source.

5

use [http://](#) or [https://](#)

Language *
The language(s) in which the resource was originally published.

2

English

3 Add Language +

- In the **Marketing** step of the wizard, you must provide a non-technical **Description** and the **Language** of your Service Catalogue - (1), (2)
- You can add more languages by clicking the **Add Language** button - (3)
- You may also optionally provide a **Tagline** (i.e., a short catchphrase describing the Service Catalogue) and a URL pointing to the **Logo** of the Service Catalogue (if available). The logo can be of any image type (e.g., PNG, JPEG) - (4), (5)

2.4 Classification

European Commission

EOSC EU Node

Resource Hub

Test Node 23/7/2025 node

Logout

Overview

Resources

Profile

Profile History

QUICKLINKS

Helpdesk

EOSC EU Node

Create a Data Source

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

Basic *

Marketing *

Classification * (1*)

Location

Contact * (3*)

Maturity

Management * (2*)

Data Source Information * (3*)

Data Source Metadata * (10*)

Acknowledgement and Acceptance

Overview

2/7 required steps completed
20%

Cancel and Exit

Save Data Source as Draft

Scientific Categorisation *

Scientific Domain *
The scientific discipline of the Data Source.

Generic

Scientific Subdomain *
The scientific sub-discipline of the Data Source.

The "Scientific Subdomain" field is required

Add Scientific Categorisation +

Tags
Keywords associated to the Data Source.

Add Tags +

Previous

Next >

- Next, in the **Classification** part of the wizard, you must select the **Scientific Domain** and **Subdomain** of your Service Catalogue. In case no scientific domain and/or subdomain are applicable to your Service Catalogue, you can select the 'Other' or 'Generic' value. You may add more domains by clicking on the **Add Scientific Categorization** button - (1), (2), (3)
- You can optionally add one more **Tags** (i.e., related keywords) to the Service Catalogue - (4), (5)

2.5 Location

Create a Data Source

Fields with (*) are mandatory.
Important: All information must be provided in English language.
Non-English submissions will result in rejection.

1

2

Geographic Locations

List of geographic locations covered by the Data Source.

4 Location

5 Contact * 3*

6 Maturity

7 Management * 2*

8 Data Source Information * 3*

9 Data Source Metadata * 10*

10 Acknowledgement and Acceptance

11 Overview

3/7 required steps completed
25%

Cancel and Exit Save Data Source as Draft

Previous Next >

- Next, you may optionally select one or more **Geographic Locations** that are covered by the Service Catalogue - (1), (2)

2.6 Contact

Create a Data Source

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

Progress: 3/7 required steps completed (25%)

Main Contact *
Important: The main contact should be a registered EOSC EU Node user. Failure to comply may result in rejection.

First Name *
The name of the main contact person for the Data Source.
Up to 20 characters

Last Name *
The last name of the main contact person for the Data Source.
Up to 20 characters

Email *
The email of the main contact person for the Data Source.

Phone Country Code
The phone country code.

Phone *
The phone of the main contact person for the Data Source.
Up to 10 characters

Organisation
The organisation of the main contact person for the Data Source.
Up to 50 characters

Position
The organisation of the main contact person for the Data Source.
Up to 20 characters

Buttons: Cancel and Exit, Save Data Source as Draft, Previous, Next >

- In the **Contact** step of the wizard, you must provide details of the main contact person for the Service Catalogue. Please note that the main contact should be a registered EOSC EU Node user
- Specifically, you must provide the main contact's **First Name**, **Last Name**, and **Email** - (1), (2), (3)
- You can optionally provide his/her **Phone**, **Organisation** and **Position** in that organization - (4), (5), (6)

2.7 Maturity

Create a Data Source

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

- Basic *
- Marketing *
- Classification *
- 4 Location
- Contact *
- 6 Maturity**
- 7 Management * (2*)
- 8 Data Source Information * (3*)
- 9 Data Source Metadata * (10*)
- 10 Acknowledgement and Acceptance
- 11 Overview

4/7 required steps completed
37%

Cancel and Exit Save Data Source as Draft

Certifications
List of certifications obtained for the Data Source (including the certification body of URL, if available).

Up to 100 characters.

1 Add Certifications + 2

Previous Next >

- Next, in the **Maturity** step, you can optionally select one or **Certifications** that were obtained for the Service Catalogue that is being onboarded - (1), (2)

2.8 Management (I/II)

Create a Data Source

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

Progress: 4/7 required steps completed (37%)

Management * (2*)

Helpdesk Page
The Helpdesk page of the Data Source.
3

Helpdesk Email *
The email of the Data Source's Helpdesk.
1

Security Contact Email *
The email of the Data Source's security contact.
2

User Manual
Link to the Data Source's user manual or documentation.
4

Terms of Use
The page containing the Terms of User for the Data Source.
5

Privacy Policy
Link to the privacy policy of the Data Source.
6

Access Policy (Acceptable Use Policy)
Link to the acceptable use policy of the Data Source.

Buttons: Cancel and Exit, Save Data Source as Draft, Previous, Next >

- In the **Management** step of the wizard, you will be asked to provide the **Helpdesk Email** and **Security Contact Email** for your Service Catalogue - (1), (2)
- You can also optionally provide the URL of the Data Source's **Helpdesk Page**, a URL pointing to the **User Manual** of the Service Catalogue, a URL pointing to the **Terms of Use** of the Service Catalogue, a URL pointing to the **Privacy Policy** of the Service Catalogue and more (see the next page) - (3), (4), (5), (6)

2.9 Management (II/II)

European Commission

EOSC EU Node

Resource Hub

Test Node 23/7/2025 node

Logout

Overview

Resources

Profile

Profile History

QUICKLINKS

Helpdesk

EOSC EU Node

Create a Data Source

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

Basic *

Marketing *

Classification *

Location

Contact *

Maturity

Management * 2*

Data Source Information * 3*

Data Source Metadata * 10*

Acknowledgement and Acceptance

Overview

4/7 required steps completed
37%

Cancel and Exit

Save Data Source as Draft

Previous

Next >

The *Helpdesk Email* field is required

Security Contact Email *

The email of the Data Source's security contact.

The *Security Contact Email* field is required

User Manual

Link to the Data Source's user manual or documentation.

use http:// or https://

Terms of Use

The page containing the Terms of User for the Data Source.

use http:// or https://

Privacy Policy

Link to the privacy policy of the Data Source.

use http:// or https://

Access Policy (Acceptable Use Policy)

Link to the acceptable use policy of the Data Source.

use http:// or https://

User Access Policy (URL)

Link to the user access policy of the Data Source.

use http:// or https://

- After scrolling down the page in the **Management** step of the wizard, you will be able to optionally provide URLs for the **Acceptable Use Policy** and **User Access Policy** of the Service Catalogue - (1), (2)

2.10 Data Source Information

Create a Data Source

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

1 Basic * 2*

2 Marketing * 2*

3 Classification * 2*

4 Location

5 Contact * 3*

6 Maturity

7 Management * 2*

✓ Data Source Information * ✓

9 Data Source Metadata * 4*

10 Acknowledgement and Acceptance * 4*

11 Overview

1/8 required steps completed
17%

Cancel and Exit Save Data Source as Draft

Datasource Type *

Software repositories

Submission Policy URL
Link to the submission policy of the Data Source.
use only https://

Preservation Policy URL
Link to the preservation policy of the Data Source.
use only https://

Version Control
Whether version control will be applied for the Data Source or not.
☐ No ☐ Yes

Persistent Identity Systems *

Persistent Identity Entity Type *
Organizations

Persistent Identity Entity Type Schemes *
DOI

Add Persistent Identity System +

Previous Next >

- Next, in the **Data Source Information** step of the wizard, you must provide the Data Source Type (i.e., **Software Repositories**, **Publications** or **Data**) and select one or more Persistent Identity Systems for the Data Source - (1), (2), (3), (4)
- You can also optionally provide the **Submission Policy URL**, the **Preservation Policy URL** of the Data Source, and indicate whether **Version Control** will be applied to the Data Source - (5), (6), (7)

2.11 Data Source Metadata (I/IV)

Create a Data Source

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

1 Basic * (2*)
2 Marketing * (2*)
3 Classification * (2*)
4 Location
5 Contact * (3*)
6 Maturity
7 Management * (2*)
8 Data Source Information * (✓)
9 **Data Source Metadata *** (4*)
10 Acknowledgement and Acceptance * (4*)
11 Overview

1/8 required steps completed
17%

Cancel and Exit Save Data Source as Draft Previous Next >

Jurisdiction *
The jurisdiction of the Data Source. (1)

Datasource Classification *
The data classification for the Data Source. (2)
Service Catalogue

Thematic *
Whether the Data Source is thematic, or not. (3)
☐ No ☐ Yes

Research Product Access Policy
The access policy of the research product related to the Data Source. (4)
Add Research Product Access Policy + (5)

Metadata Licensing of Research Products

Metadata License Name (6)
The license name of the research product metadata related to the Data Source.

Metadata License URL (7)
The license URL of the research product metadata related to the Data Source.
use only https://

- In the **Data Source Metadata** step of the wizard, you must initially select the **Data Source Classification**, which should be set to **Service Catalogue**. Then, you must fill-in the rest of the mandatory fields for Service Catalogues, which include the **Jurisdiction** (i.e., pool of users the data source is intended to serve) of the Service Catalogue (either **Global**, **Institution**, **National**, or **Regional** aka multiple countries) and whether the Service Catalogue is **Thematic** or not (i.e., organized around a specific theme, subject, or domain of interest, rather than being general-purpose or broadly scoped) - (1), (2), (3)
- You can optionally select one or more **Access Policies** for the research product related to the Service Catalogue that is being onboarded - (4), (5)
- You may also optionally provide details for the related **Research Product Metadata Licensing** (i.e., name and URL) - (6), (7)

2.12 Data Source Metadata (II/IV)

Create a Data Source

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

1 Basic * (2*)
2 Marketing * (2*)
3 Classification * (2*)
4 Location
5 Contact * (3*)
6 Maturity
7 Management * (2*)
8 Data Source Information * (✓)
9 **Data Source Metadata *** (4*)
10 Acknowledgement and Acceptance * (4*)
11 Overview

1/8 required steps completed
17%

Cancel and Exit Save Data Source as Draft Previous Next >

Research Product Metadata Access Policies
Access policies related to the research product metadata.
3
Add Research Product Metadata Access Policy +

Harvestable
Whether the Data Source will be harvestable by other entities, or not.
4
☐ No ☐ Yes

Data Source Access URL *
The Base URL of the Data Source.
1
use only https://

Base URL Protocol *
2

OAI Sets
This is the OAI Sets.
5
Add OAI Set + 6

Format
7

- Scrolling down the **Data Source Metadata** step of the wizard, you will be asked to provide the **Base URL** of the Service Catalogue and its protocol (FTP, OAI, Rest API or Other) - (1), (2)
- You may also optionally provide one or more **Access Policies** for the related research product metadata, indicate whether the Service Catalogue is harvestable by other entities or not, add one or more **OAI Sets** and select the **Format** of the Service Catalogue - (3), (4), (5), (6), (7)

2.13 Data Source Metadata (III/IV)

Create a Data Source

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

- 1 Basic *
- 2 Marketing *
- 3 Classification *
- 4 Location
- 5 Contact *
- 6 Maturity
- 7 Management *
- 8 Data Source Information *
- 9 **Data Source Metadata ***
- 10 Acknowledgement and Acceptance *
- 11 Overview

1/8 required steps completed
17%

Cancel and Exit Save Data Source as Draft

Compatibility
Declare whether the Data Source is compatible with OpenAIRE guidelines. Compatibility with OpenAIRE guidelines is mandatory only for Research Product Catalogues.

1

Compliance to OpenAIRE Guidelines
I acknowledge that all appropriate steps have been performed in OpenAIRE provide for this datasource to be compliant with the latest OpenAIRE Guidelines "OpenAIRE for institutional and Thematic Repositories (OpenAIRE 4.0)". Note that compliance is mandatory only for Research Product Catalogues. In case the data source is of a different type you can select the "No" value.

2

☐ No ☐ Yes

Repository Identifier
The repository identifier of the Data Source.

3

• Make sure your Repository is already registered in one of the following global registries - [OpenDOAR](#) (for Literature Repositories), [Re3Data](#) (for Data Repositories) or [FAIRsharing](#) (for Data and Literature Repositories), and is compatible with the [OpenAIRE Guidelines](#)
• <https://www.ocemains.eu/register-your-data-repository-in-ocemains>
• <https://www.ocemains.eu/validator-registration-guide>

Repository Identifier Type
The repository identifier type (e.g. opendoar, re3data, fairsharing) of the Data Source.

4

Alternative Identifiers
Alternative Identifier
This is Alternative Identifier

• Make sure your Repository is already registered in one of the following global registries - [OpenDOAR](#) (for Literature Repositories), [Re3Data](#) (for Data Repositories) or [FAIRsharing](#) (for Data and Literature Repositories), and is compatible with the [OpenAIRE Guidelines](#)

Previous Next >

- Scrolling further down the **Data Source Metadata** step of the wizard, you may optionally declare whether the Service Catalogue is compatible with the [OpenAIRE guidelines 4.0](#) - (1), (2)
- You may also provide the **Repository Identifier** (and its type) of the Service Catalogue. The repository must be already registered in one of the following global registries: [OpenDOAR](#) (for Literature Repositories), [Re3Data](#) (for Data Repositories) or [FAIRsharing](#) (for Data and Literature Repositories). The repository must also be compatible with the [OpenAIRE Guidelines 4.0](#) - (3), (4)

2.14 Data Source Metadata (IV/IV)

European Commission

EOSC EU Node

Resource Hub

Test Node 23/7/2025 node

Logout

Overview

Resources

Profile

Profile History

QUICKLINKS

Helpdesk

EOSC EU Node

Create a Data Source

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

- Basic *
- Marketing *
- Classification *
- 4 Location
- Contact *
- 6 Maturity
- Management *
- Data Source Information *
- 9 Data Source Metadata *** 10*
- 10 Acknowledgement and Acceptance
- 11 Overview

6/7 required steps completed
58%

Cancel and Exit Save Data Source as Draft

Guidelines "OpenAIRE for institutional and Thematic Repositories (OpenAIRE 4.0).
☐ No ☐ Yes

Repository Identifier *
The repository identifier of the Data Source.

Repository Identifier Type *
The repository identifier type (e.g. opendoar, re3data, fairsharing) of the Data Source.

Alternative Identifiers

Alternative Identifier
This is Alternative Identifier

Alternative Identifier Type
This is Alternative Identifier Type

Add Alternative Identifier +

Previous Next >

- Finally, scrolling at the bottom of the Data Source Metadata step of the wizard, you can optionally provide one or more alternative repository identifiers and their type - (1), (2), (3)

2.15 Acknowledgment and Acceptance

Create a Data Source

Fields with (*) are mandatory.
Important: All information must be provided in English language.
Non-English submissions will result in rejection.

1 Basic *
2 Marketing *
3 Classification *
4 Location
6 Contact *
6 Maturity
Management *
Data Source Information *
Data Source Metadata *
10 Acknowledgement and Acceptance
11 Overview

7/7 required steps completed
100%

Cancel and Exit Save Data Source as Draft

Acknowledgement and Acceptance *
I confirm that I accept the following conditions:

- ☐ Information submitted in the present form will be automatically stored in the EOSC EU Node Catalogue services.
- ☐ Selected information from the present form will be automatically presented in the EOSC EU Node Resource Hub upon successful Onboarding.
- ☐ Registered service endpoints will be automatically monitored by the EOSC EU Node Monitoring Service for basic service uptime checks.
- ☐ The registered security contact point may be contacted by the EOSC EU Node Security Team in case of security incident.

Previous Next >

- In the **Acknowledgment and Acceptance** step of the wizard, you must provide your consent to the following:
- Information submitted in the present form to be automatically stored in the EOSC EU Node Catalogue services - (1)
- Selected information from the present form to be automatically presented in the EOSC EU Node Resource Hub upon successful Onboarding - (2)
- Registered service endpoints to be automatically monitored by the EOSC EU Node Monitoring Service for basic service uptime checks - (3)
- The registered security contact point to be contacted by the EOSC EU Node Security Team in case of security incident - (4)

2.16 Overview

EOSC EU Node

Resource Hub

Test Node 23/7/2025 node

Logout

Overview

Resources

Profile

Profile History

QUICKLINKS

Helpdesk

EOSC EU Node

Create a Data Source

Fields with (*) are mandatory.
Important: All information must be provided in the English language.
Non-English submissions will result in rejection.

Basic *

Marketing *

Classification *

Location

Contact *

Maturity

Management *

Data Source Information *

Data Source Metadata *

Acknowledgement and Acceptance

Overview

7/7 required steps completed 100%

Cancel and Exit

Save Data Source as Draft

Basic

Name: Test DS 23/7/2025 No2

Abbreviation:

Webpage: https://test.com

Marketing

Description: Test

Tagline:

Logo:

Languages: English

Classification

Scientific Categorisation:

Scientific Domain: Generic

Scientific Subdomain: Generic

Tags:

Location

Geographic Locations:

Contact

Main Contact:

Previous

Publish Data Source

- Finally, at the **Overview** step of the wizard, you have the opportunity to go through all the details that you have provided for the Data Source to be onboarded
- To edit a field before submitting, you may either click on the corresponding step title in the left pane of the wizard, or on the **Edit** button above each category - (1), (2)
- To submit the application for review, click on the **Publish Data Source** button. You will be presented with a confirmation modal. Upon successful submission the Data Source's application will appear as having the 'Pending' status in the **Resources** tab in your **Node Dashboard**. Once the application is validated and accepted by the EOSC EU Node Onboarding Team and the European Commission, your Data Source will appear as 'Onboarded' in the **Resources** tab in your **Node Dashboard** - (3)

Appendix D: Implementation options

Service Catalogues typically stem from specific requirements, such as the marketing of services, and entail costs associated with their implementation, maintenance, and operation. EOSC Nodes may therefore choose not to provide Service Catalogues if they are not considered necessary by their user community. For example, this is the case where a Node offers a fixed set of services accessible through a directory on a static portal. When a service catalogue is not available, it is important for the EOSC Node to evaluate whether implementing one is feasible and beneficial, taking into account any applicable requirements and resource availability (the EOSC Node charters may have already indicated a roadmap).

To establish a service catalogue, EOSC Nodes have the option to:

Develop new software. Develop a software application from scratch whose export APIs and metadata format match the requirements specified in Appendix A and B.

Reuse and deploy existing open-source software. Two EEN-compatible solutions are available from the EEN, which can offer its service catalogue software, the EOSC Beyond project, and the EOSC CSC Node (<https://github.com/CSCfi/EOSCnode>).

Request an aaS option. Two solutions are available: the first from the EEN, which can offer its Service Resource Catalogue as a Service; the second option from the EOSC IT Node and the EOSC BlueCloud Node, which can offer a service catalogue aaS via [D4Science.org](https://d4science.org).